

Interview AI

AI-Powered Technical Interview Platform

Product Overview & Walkthrough

Generated: 20 May 2026

Table of Contents

1. How It Works — Overview

2. Candidate Flow

- 2.1 Landing Page
- 2.2 Resume Upload
- 2.3 Role Detection & Confirmation
- 2.4 Video Interview
- 2.5 Completion

3. Admin Flow

- 3.1 Admin Login
- 3.2 Sessions Dashboard
- 3.3 Candidate Detail & Analysis
- 3.4 Settings (API Key Management)

4. AI Analysis Process

5. Technical Architecture

1. How It Works — Overview

Interview AI is an AI-powered technical interview platform that automates the screening process for tech candidates. It is designed as an analysis tool, not a filter — the goal is to understand a candidate's strengths, not eliminate them.

The platform uses Anthropic's Claude AI to:

- Detect the best-fit technical role from a candidate's resume
- Generate 6 adaptive, role-specific interview questions
- Analyse each answer for technical accuracy, communication, and depth
- Produce a detailed, downloadable PDF report for admin review

The entire flow takes approximately 15–20 minutes for a candidate.

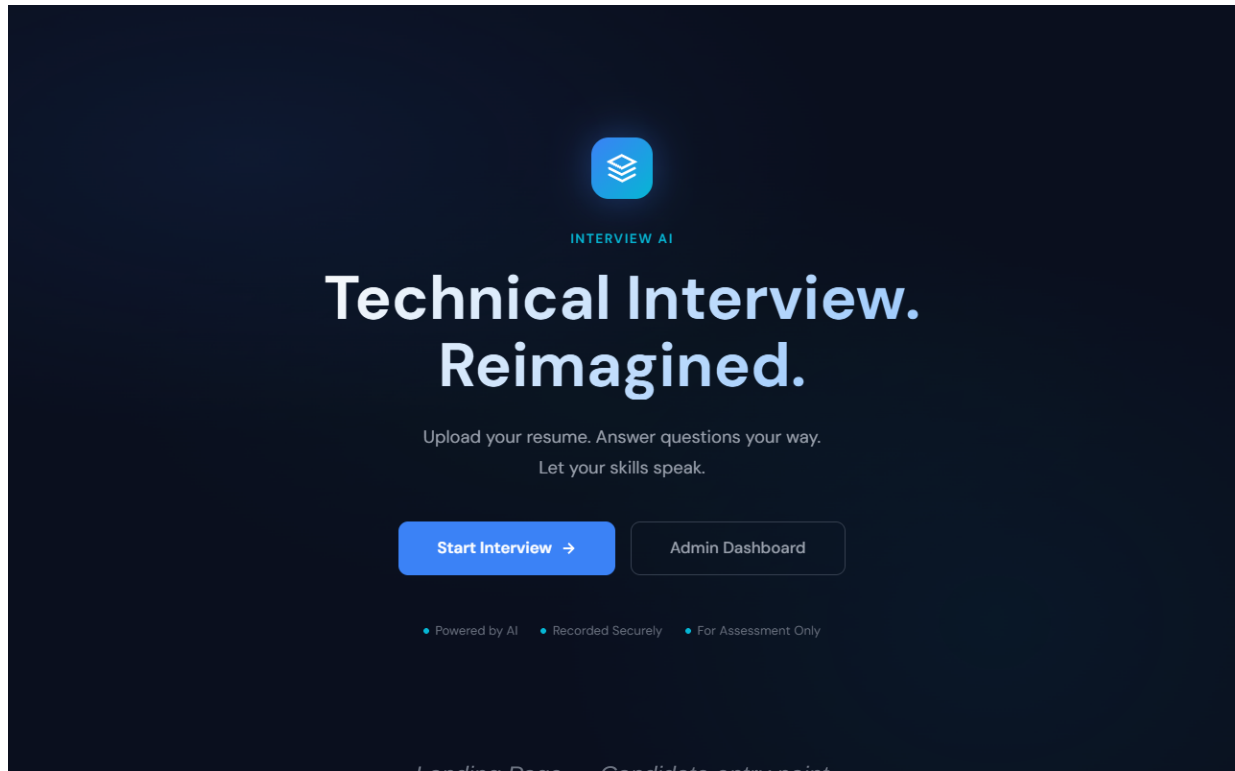
End-to-End Flow

1. Candidate uploads resume (PDF, DOCX, or TXT)
2. AI analyses resume and detects the best-fit tech role
3. Candidate confirms or changes the role
4. AI generates 6 tailored interview questions
5. Candidate answers each question via webcam + microphone
6. Browser captures video recording + live speech-to-text transcript
7. AI scores and analyses each answer individually
8. AI generates a comprehensive final assessment
9. PDF report is generated and available for admin download
10. Admin reviews all sessions, scores, and reports in the dashboard

2. Candidate Flow

2.1 Landing Page

The landing page is the first thing candidates see. It features a clean, dark-themed design with a bold headline: "Technical Interview. Reimagined." Two call-to-action buttons let users either start an interview or access the admin dashboard. Trust indicators at the bottom reassure candidates that the platform is AI-powered, recorded securely, and for assessment purposes only.



Landing Page — Candidate entry point

2.2 Resume Upload

Candidates enter their full name and email, then upload a resume via drag-and-drop or file picker. Accepted formats are PDF, DOC, DOCX, and TXT (max 5 MB). After uploading, the AI analyses the resume text and detects skills, experience level, and the best-fit technical role.

Upload Your Resume

We'll analyse your background and tailor the interview to you.

Full Name

Jane Smith

Email Address

jane@example.com

Drag & drop your resume here

PDF, DOC, DOCX or TXT - max 5 MB

Analyse & Continue →

Upload Page — Resume upload with drag-and-drop zone

2.3 Role Detection & Confirmation

After upload, the AI shows its detected role (e.g. "Frontend Developer") with a confidence percentage, a list of detected skills as pill tags, and a summary of the candidate's background. The candidate can accept the detected role or override it using a dropdown menu with 14 supported tech roles.

Supported Roles:

Frontend Developer • Backend Developer • Full Stack Developer • DevOps Engineer • Cybersecurity Analyst • UI/UX Designer • Mobile Developer • Data Engineer • ML/AI Engineer • SAP Consultant • Cloud Architect • QA Engineer • Blockchain Developer • Embedded Systems Engineer

2.4 Video Interview

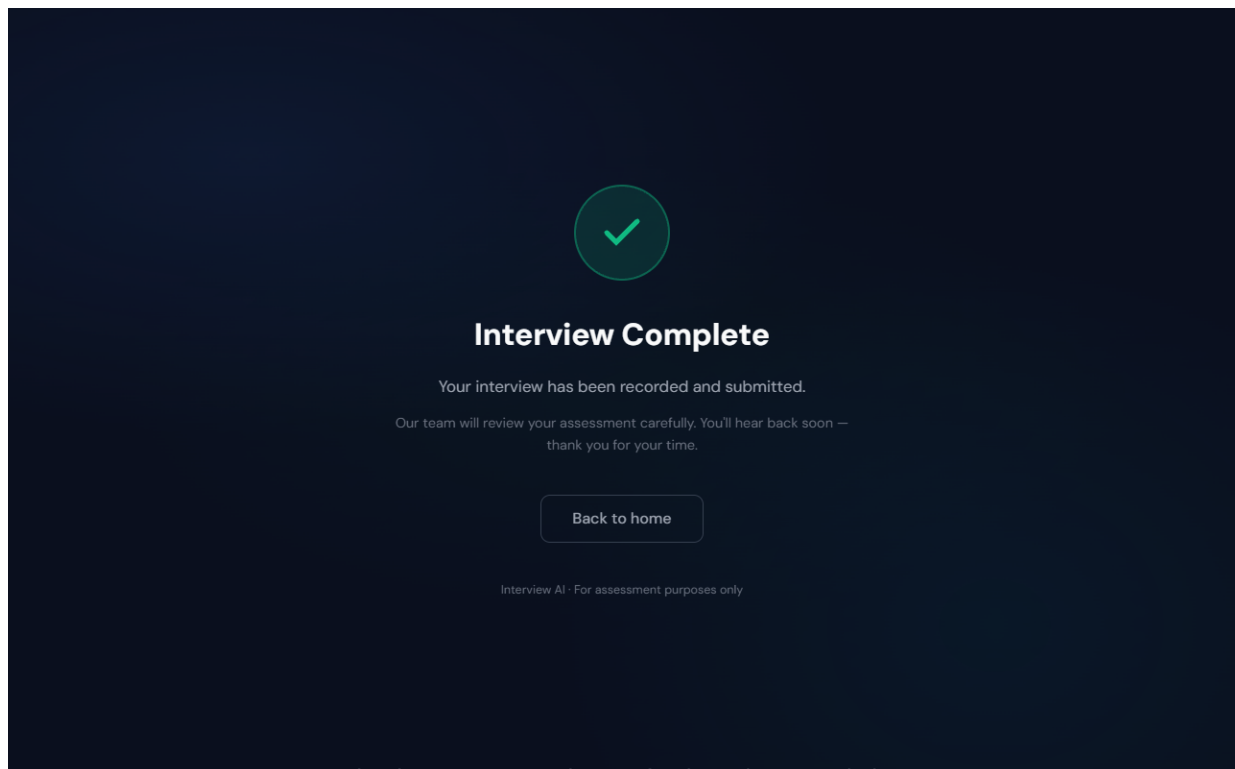
The interview screen is split into two columns: the left shows the live webcam feed with a recording indicator, and the right displays the current question with competency/difficulty tags and a circular countdown timer.

For each question:

- A 5-second "prepare" countdown shows the question before recording starts
- Recording begins automatically (video + audio via MediaRecorder API)
- Speech recognition runs in parallel, showing a live ghost transcript on screen
- The countdown timer warns the candidate at 30 seconds remaining
- When time expires (or the candidate clicks "Stop & Submit"), recording stops
- The video blob and transcript are sent to the server
- AI analyses the answer before moving to the next question

2.5 Completion

After all questions are answered, the candidate sees a confirmation page with an animated checkmark. No scores or analysis are shown to the candidate — the page simply confirms their interview was recorded and submitted, and that the team will review it.

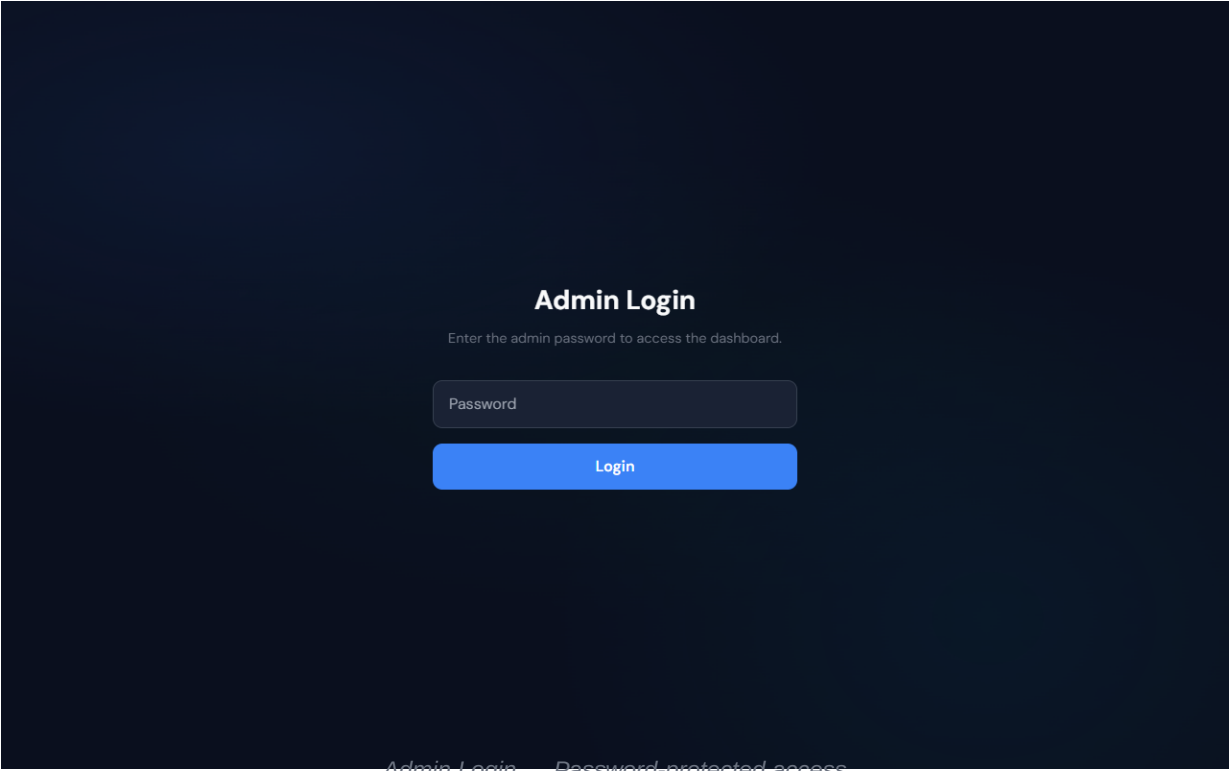


Thank You Page — Shown after interview completion

3. Admin Flow

3.1 Admin Login

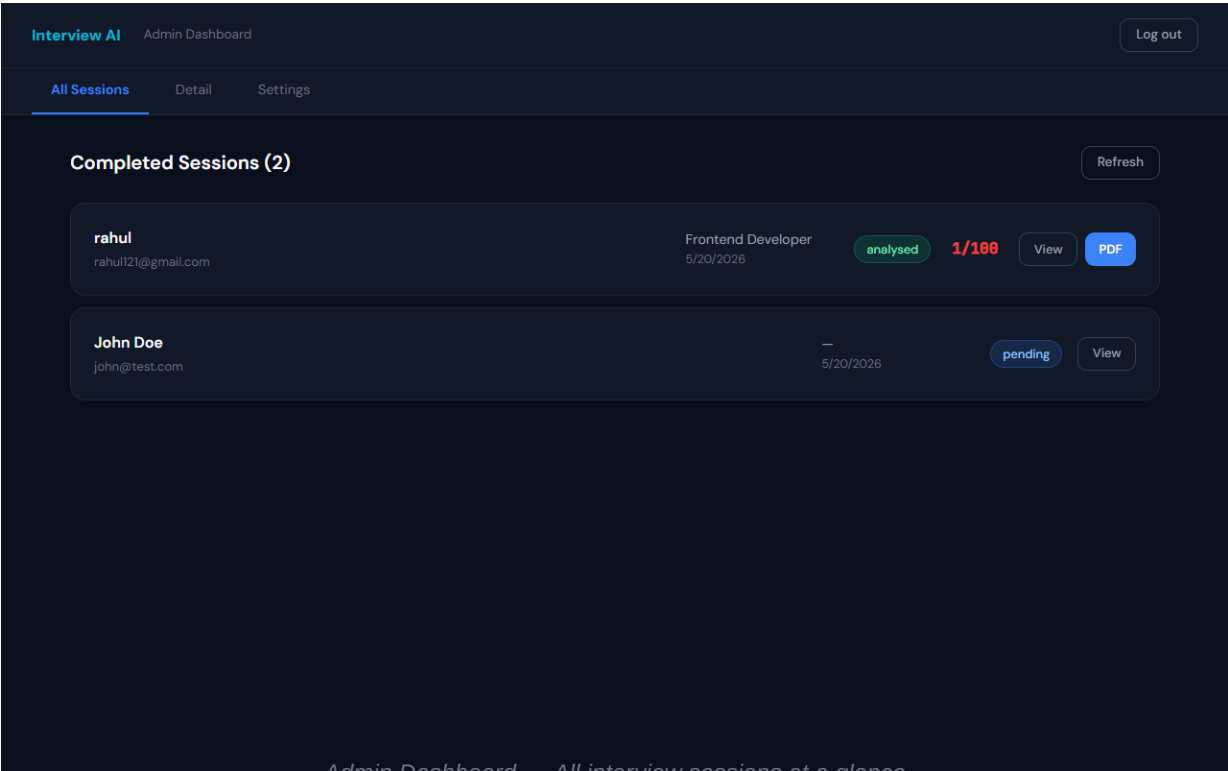
The admin dashboard is protected by a simple password gate. The default password is "admin123" (configurable in the .env file). After logging in, the admin has access to all interview sessions.



Admin Login — Password-protected access

3.2 Sessions Dashboard

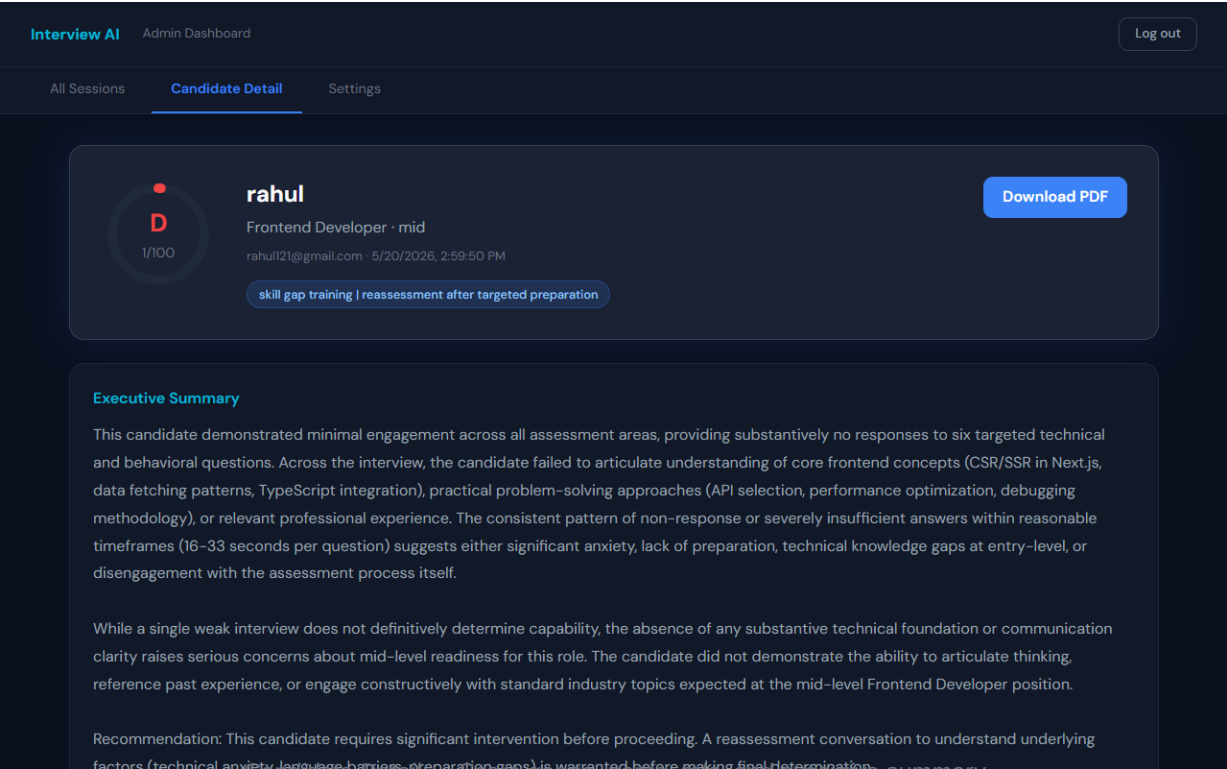
The sessions list shows all completed interviews with the candidate's name, email, detected role, date, status badge (pending / in progress / analysed), and overall score. Each row has a "View" button for the full analysis and a "PDF" button to download the generated report.



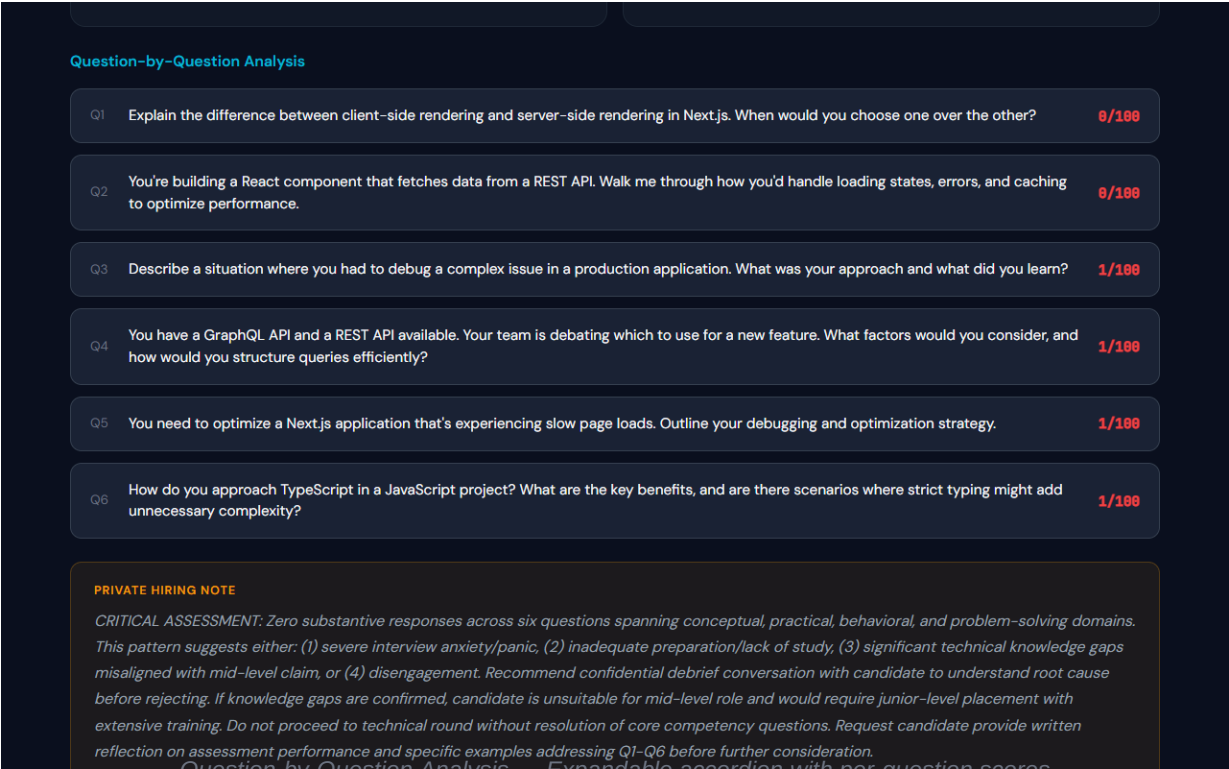
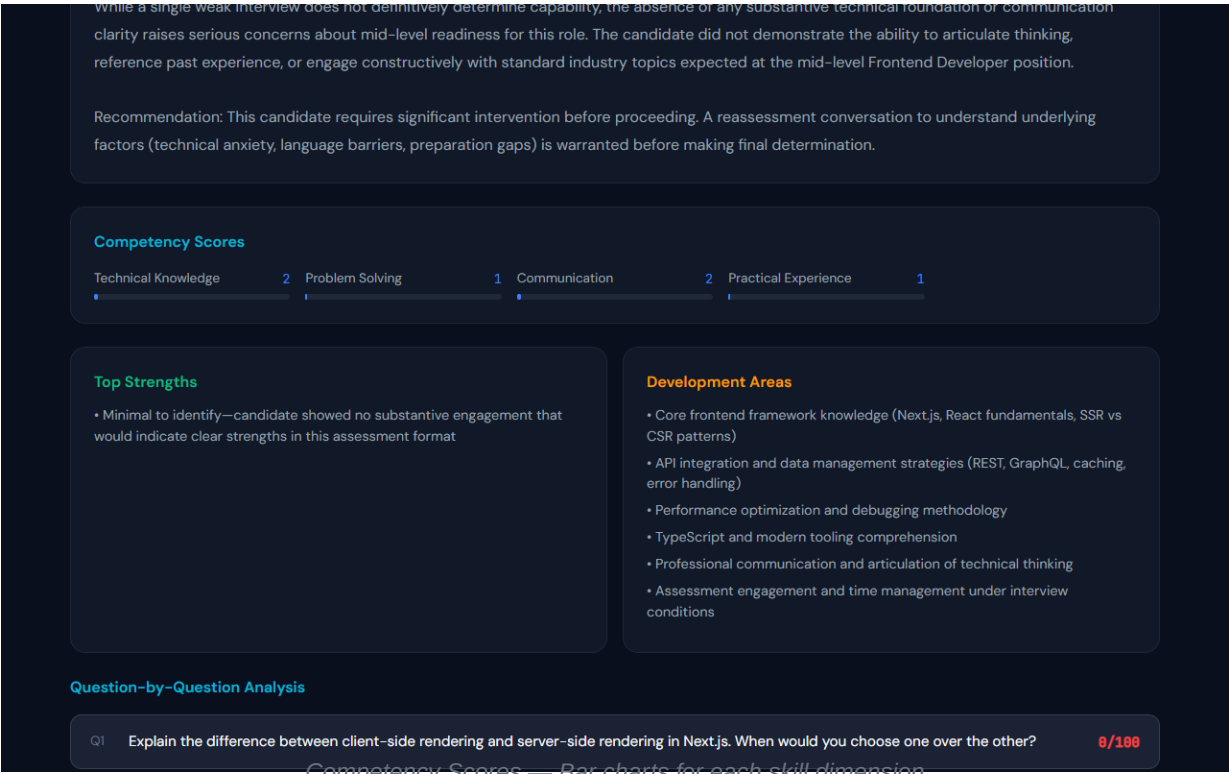
3.3 Candidate Detail & Analysis

Clicking "View" opens the full analysis for a candidate. This includes:

- Score Gauge — Overall score out of 100 with a letter grade (A/B/C/D)
- Executive Summary — A 3–4 paragraph AI-written assessment
- Recommended Next Step — e.g. "proceed to technical round" or "skill gap training"
- Competency Scores — Bar charts for Technical Knowledge, Problem Solving, Communication, and Practical Experience (each 0–100)
- Top Strengths & Development Areas — Listed side by side
- Question-by-Question Accordion — Each question expandable with sub-scores, strengths, areas to explore, answer summary, and evaluator note
- Private Hiring Note — Honest internal note only visible to admins

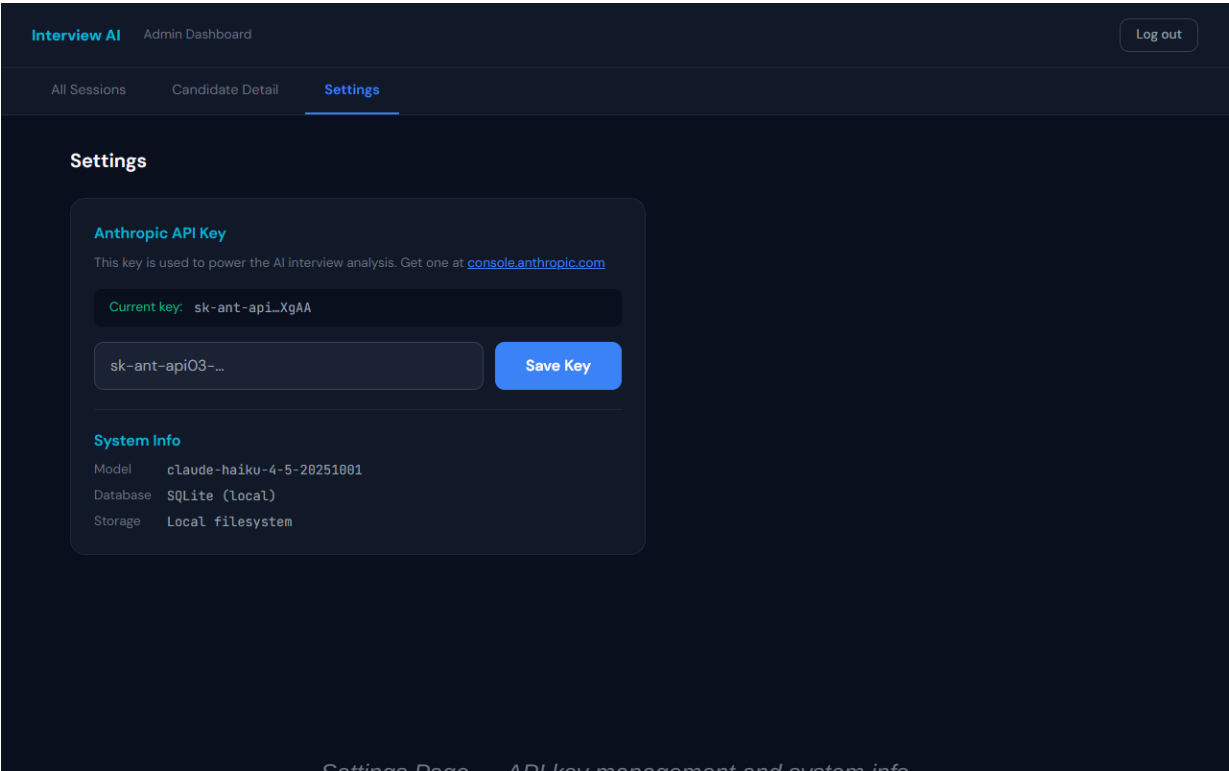


Candidate Detail — Score gauge, name, role, and executive summary



3.4 Settings (API Key Management)

The Settings tab lets admins update the Anthropic API key directly from the dashboard — no need to edit config files. It shows the current key (masked for security), provides an input to enter a new key, and displays system info (model name, database type, storage type).



Settings Page — API key management and system info

4. AI Analysis Process

The platform makes four distinct AI calls during each interview session:

4.1 Role Detection

Input: Full resume text extracted from the uploaded file.

Output: Primary role (from 14 options), confidence score (0–1), detected skills, experience level (junior/mid/senior/lead), a 2–3 sentence summary, and secondary role suggestions.

4.2 Question Generation

Input: Confirmed role, detected skills, and experience level.

Output: 6 questions with a mix of competency areas — 1–2 conceptual/theory, 2–3 practical/scenario, 1 problem-solving, and 1 behavioural. Each question includes difficulty level, time limit, expected keywords, and a follow-up hint (never shown to the candidate).

4.3 Per-Answer Analysis

Input: The question, speech-to-text transcript, role, and time taken.

Output: Score 0–100 with a four-part breakdown (technical accuracy, communication clarity, depth of knowledge, practical application — each 0–25), keywords mentioned vs missed, strengths, areas to explore, a neutral answer summary, and a constructive evaluator note.

4.4 Final Session Analysis

Input: All question summaries and individual scores.

Output: Overall score 0–100, letter grade (A/B/C/D), role fit percentage, a multi-paragraph executive summary, competency scores across four dimensions, top strengths, development areas, recommended next step, alternative role suggestions, and a private hiring note.

5. Technical Architecture

Frontend: React 18 + Vite, Tailwind CSS, React Router

Backend: Node.js + Express, RESTful API on port 3001

AI Engine: Anthropic Claude API (claude-haiku-4-5)

Database: SQLite via better-sqlite3 (local file)

Video Capture: Browser MediaRecorder API (WebM/VP9)

Speech-to-Text: Web Speech API (browser-native)

PDF Reports: PDFKit (server-side generation)

File Storage: Local filesystem (resumes, recordings, reports)

File Structure

```
interview-agent/  
  server/  
    index.js           Express API entry point  
    services/claude.js All AI calls (role detect, questions, analysis)  
    services/parser.js Resume text extraction (PDF/DOCX/TXT)  
    services/reportGenerator.js PDF report builder  
    routes/            upload, interview, recording, analysis, admin  
    db/               SQLite schema and database module  
  client/src/  
    pages/            Landing, Upload, RoleConfirm, Interview, ThankYou, Admin  
    components/       VideoRecorder, QuestionCard, CountdownTimer, etc.  
  tests/             End-to-end test scenarios for 4 roles  
  .env               API keys and configuration
```

